

Research report of June 2026

As I did experiment of KAN on synthesized dataframes I was focused on real dataframes, the data source I chose is EPFL SMLM 2D Challenge — Bundled

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Abstract

This report presents the methods and results of five progressively refined pipelines for single-molecule localization microscopy (SMLM) based on Kolmogorov–Arnold Networks (KANs). Each version is described in terms of its detection algorithm, localization algorithm, training strategy, and inference steps. All models were trained on EPFL data and evaluated blindly on the first 200 frames of the EPFL test sequence (1,365 ground-truth activations). The final Full-Frame Dual KAN pipeline achieves 91.4% Jaccard index, 100% precision, 91.4% recall.

1. Introduction and Evaluation Protocol

SMLM reconstructs super-resolution images by detecting sparse fluorescent emitters in camera frames and estimating their sub-pixel coordinates. The project goal was to design pipelines where KANs participate in detection and/or localization, process whole 64×64-pixel frames, and adapt to a variable number of emitters per frame (1–16, EPFL-like density).

1.1 Common experimental settings

Setting	Description
Input frame size	64 × 64 pixels
Camera pixel size	100 nm
Super-resolution rendering	10 nm per pixel, Gaussian splatting
Training data	Synthetic TIFF movies with known emitter positions
Synthetic emitter density	1–16 emitters per frame (EPFL-like)
Synthetic noise model	Poisson photons, uniform background, Gaussian readout noise
Blind test	EPFL test frames only (no ground truth used during inference)

Table 1. Shared experimental and evaluation settings across all versions.

1.2 Comparative results overview

Table 2 summarizes blind-test performance for every pipeline variant.

Version	Detection method	Localization method	Predictions	Recall	Precision	Jaccard
v1 — Patch KAN	Classical intensity peaks	KAN on 11×11 patch	1,190	87.2%	100%	87.2%
v2.1 — Slot KAN	Global CNN + slot confidence	Same network (joint)	~27	≈2%	—	Failed

v2.2 — Dense Grid KAN	Per-cell KAN confidence	Per-cell KAN coordinates	1,768	84.2%	65.0%	57.9%
v2.3 — Heatmap + Patch KAN	CNN heatmap + peak finding	KAN on 11×11 patch	1,385	91.5%	90.2%	83.2%
v2.4 — Full-Frame Dual KAN	Spatial KAN heatmap + peaks	KAN on full frame + peak hint	1,248	91.4%	100%	91.4%

Table 2. EPFL blind-test comparison (200 frames, 1,365 ground-truth activations).

2. Version 1 — Classical Peak Detection with Patch KAN Localization

Objective

Establish a baseline pipeline in which detection is handled by a classical image-processing method and only localization is performed by a KAN.

Method

Detection approach:

Each frame is background-subtracted and normalized. Candidate emitters are identified as local intensity maxima whose value exceeds the frame mean plus 4.5 standard deviations. A non-maximum suppression step removes peaks closer than 3 pixels, keeping the brightest candidate when two peaks conflict.

Localization approach:

For every detected peak, an 11×11 pixel region centered on the integer peak coordinate is extracted. The patch is normalized and flattened into a feature vector. (3 layer KAN)

Network architecture (conceptual)

The localizer is a Kolmogorov–Arnold Network with three hidden stages. learnable univariate activation functions are placed on network edges (B-spline basis). Input: 11×11 patch (121 values). Detection is not part of the neural network.

Results

Metric	Synthetic validation	EPFL blind test (200 frames)
Jaccard index	93.0%	87.2%
Localization RMSMD	28.9 nm	65.8 nm
Total predictions / GT	343 / 369	1,190 / 1,365
True positives	343	1,190
False positives	0	0
False negatives	26	175

Table 3. Version 1 results.

KAN final reconstruction — all 200 EPFL frames

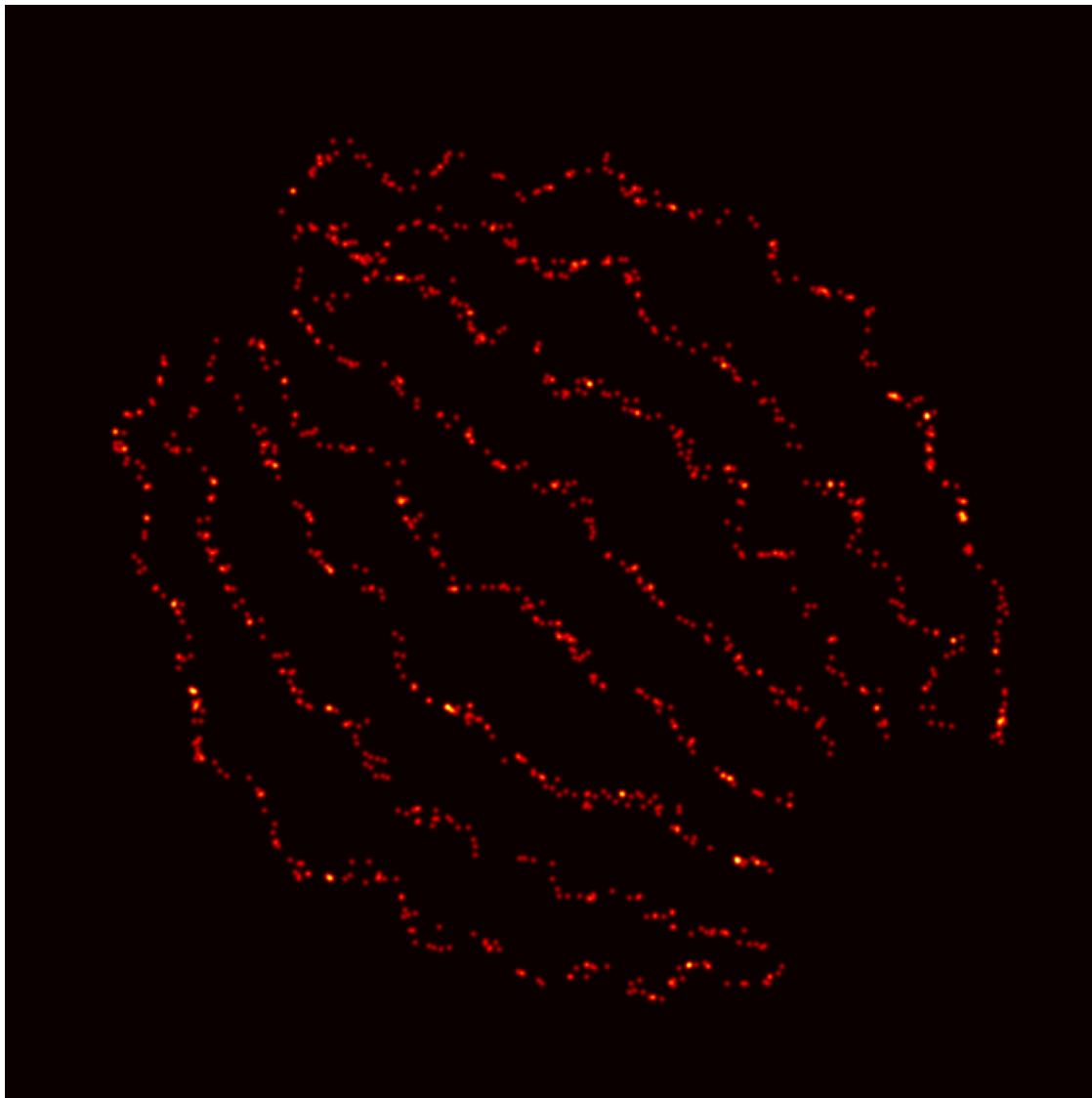


Figure 3. Version 1 — EPFL super-resolution reconstruction.

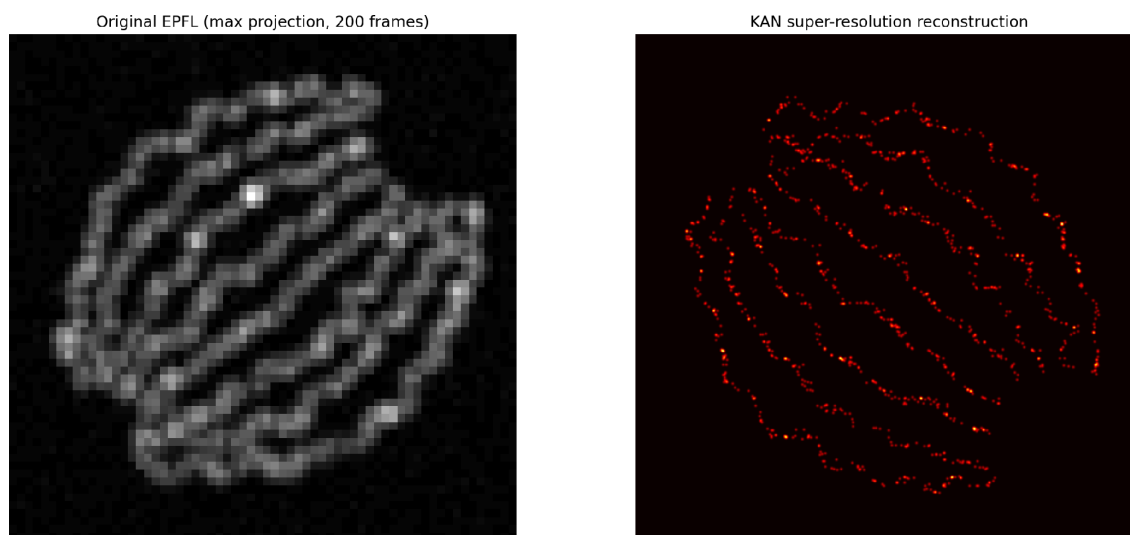


Figure 4. Version 1 — EPFL localization comparison.

Remarks

Version 1 achieves zero false positives but misses approximately 13% of true emitters, because classical peak detection is conservative on real EPFL data. Localization accuracy on synthetic data (28.9 nm RMSMD) degrades on EPFL (65.8 nm).

3. Version 2.1 — Slot-Based Frame KAN (Failed)

Objective

Replace classical detection with a neural approach that processes the entire frame in one forward pass and predicts a fixed maximum number of emitters per frame using KAN.

Method

Detection and localization (joint):

A convolutional encoder reads the full 64×64 frame and compresses it through global average pooling into a single global feature vector.

Training procedure:

Supervised training on synthetic frames with Hungarian matching: predicted slots are optimally matched to ground-truth emitters within each frame. Matched slots are trained to regress true positions and high confidence; unmatched slots are trained toward zero confidence.

Results

Metric	Observation
EPFL predictions (200 frames)	~27 (expected ~1,365)
Recall	Approximately 2% — catastrophic under-detection
Root cause	All slots share one global feature vector and cannot attend to different spatial regions

Table 4. Version 2.1 results — design failure.

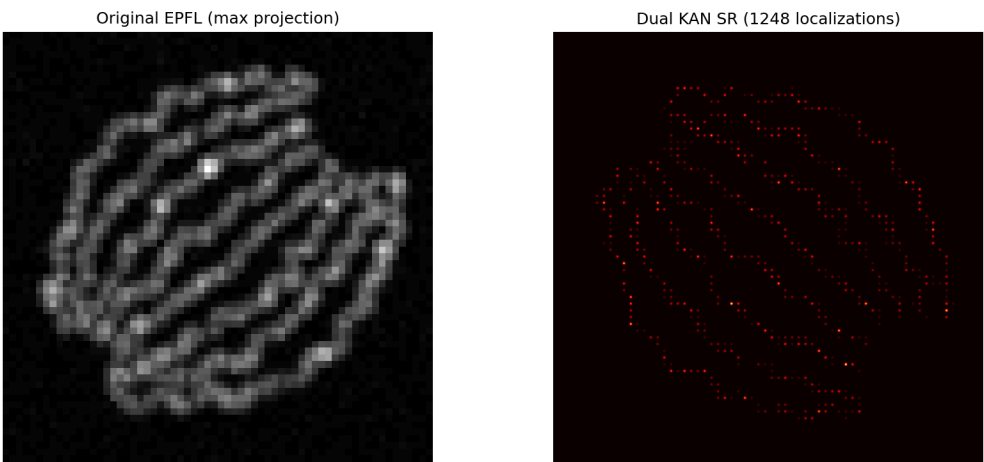


Figure 4. Version 2-1 — EPFL localization comparison.

4. Version 2.2 — Dense Spatial Grid KAN

Objective

Preserve spatial structure by dividing the frame into a grid of cells, each with its own KAN that jointly performs detection and localization within that cell.

Method

Detection and localization (per grid cell):

A spatial convolutional encoder outputs a feature map aligned with a 16×16 grid over the 64×64 frame (each cell covers 4×4 pixels). For every grid cell, a KAN receives the local feature vector and outputs three values: confidence that an emitter is present, and local sub-cell x and y coordinates. Global positions are obtained by combining cell index with local offsets.

Training procedure:

Trained on 1,000,000 synthetic frames with 1–16 emitters per frame. Each cell is supervised with binary confidence targets and coordinate regression for cells containing ground-truth emitters. The best model checkpoint was selected using a validation metric that favored high detection counts relative to ground truth.

Results

Metric	EPFL blind test
Detection recall (approx.)	84.2%
Precision	65.0%
Jaccard index	57.9%
Predictions / GT	1,768 / 1,365
False positives	619
False negatives	216
Localization RMSMD	89.9 nm

Table 5. Version 2.2 results.

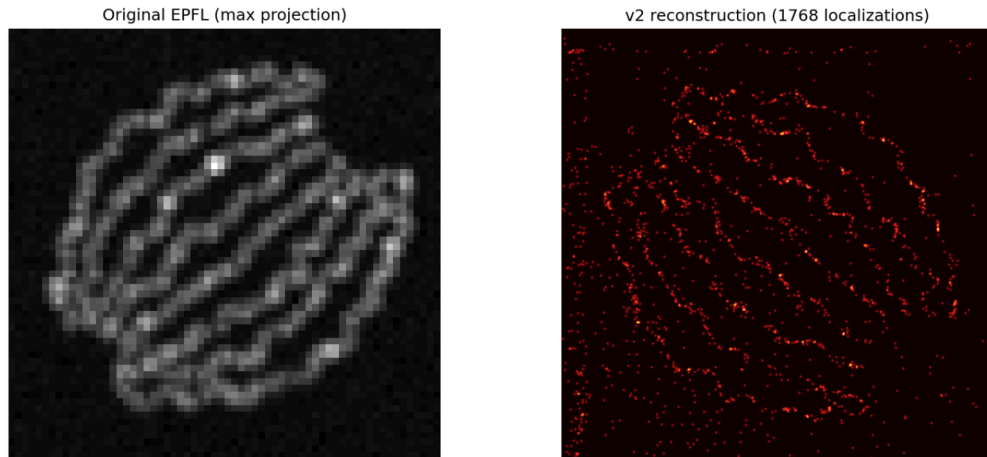


Figure 4. Version 2-2 — EPFL localization comparison.

5. Version 2.3 — Heatmap Detection with Patch KAN Localization

Objective

Decouple detection from localization: use a convolutional heatmap for robust emitter finding, and reserve the KAN exclusively for localization at detected peak locations.

Method

Detection approach:

A convolutional network maps each frame to a single-channel probability heatmap the same size as the input (64×64). During training, the target heatmap is a sum of Gaussian kernels centered on ground-truth emitter positions. At inference, local maxima of the predicted heatmap are extracted using absolute and relative thresholds followed by non-maximum suppression. The number of detected emitters varies naturally per frame.

Localization approach:

Identical to Version 1: for each heatmap peak, an 11×11 patch is extracted and passed through a three-layer KAN.

Inference algorithm

1. Preprocess the input frame.
2. Run the detection network to obtain a 64×64 heatmap.
3. Find local maxima above absolute and relative thresholds.
4. Apply non-maximum suppression (minimum 3-pixel separation).
5. For each surviving peak, extract an 11×11 patch and run the KAN localizer.
6. Combine integer peak coordinates with KAN offsets; convert to nanometres.

Results

Metric	EPFL blind test
Jaccard index	83.2%
Recall	91.5%
Precision	90.2%
Predictions / GT	1,385 / 1,365
True positives	1,249
False positives	136
False negatives	116
Localization RMSMD	52.6 nm

Table 6. Version 2.3 results.

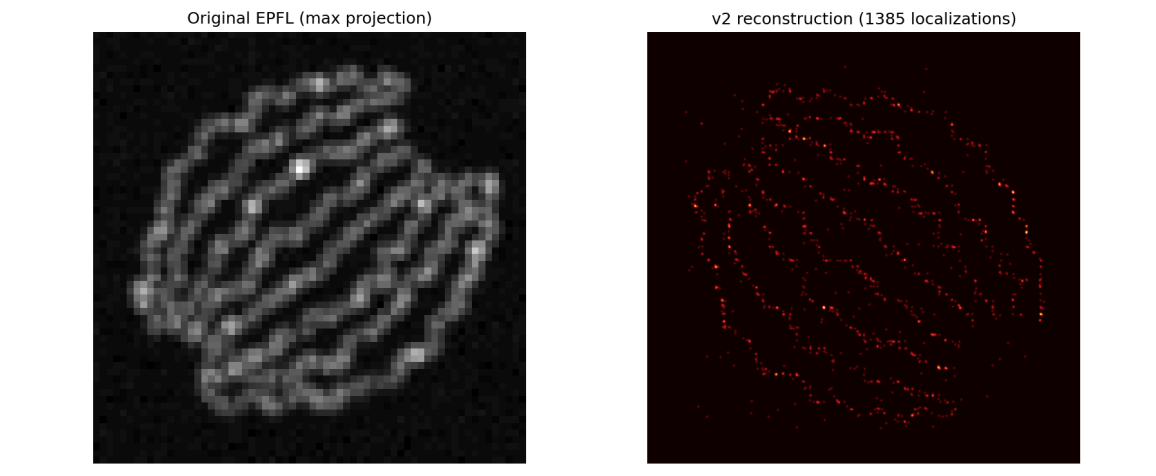


Figure 4. Version 2-3 — EPFL localization comparison.

6. Version 2.4 — Full-Frame Dual KAN (Final Model)

Objective

KAN-based detection, KAN-based localization with full-frame input, and variable emitter count per frame while improving accuracy over all prior versions.

Method

Detection approach:

A full-resolution convolutional encoder preserves the 64×64 spatial layout and produces a 128-channel feature map. A KAN is applied independently to the feature vector at every pixel location, producing a scalar detection score per pixel. These scores form a heatmap; after sigmoid normalization, peaks are extracted with non-maximum suppression. This combines the calibrated detection of Version 2.3 with KAN-based spatial scoring.

Localization approach:

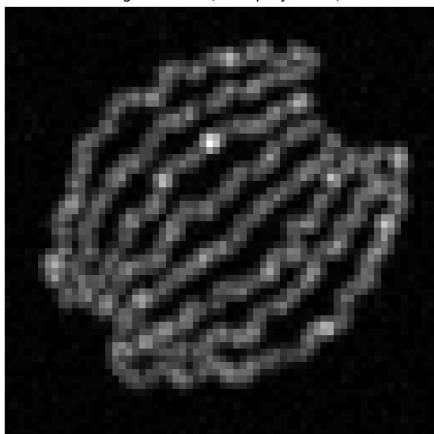
For each detected peak, the entire frame (all 4,096 pixel values) is concatenated with the normalized peak coordinates and fed into a separate KAN. Using the full frame gives the localizer global context (neighbouring emitters, background structure, PSF shape).

Results

Metric	Synthetic validation	EPFL blind test
Jaccard index	93.0%	91.4%
Recall	—	91.4% (1,248 / 1,365)
Precision	100% (zero FP)	100% (zero FP)
Predictions / GT	—	1,248 / 1,365
True positives	3,826 (val set total)	1,248
False positives	0	0
False negatives	287 (val set total)	117
Localization RMSMD	—	43.7 nm

Table 7. Version 2.4 results — final model.

Original EPFL (max projection)



Dual KAN SR (1248 localizations)

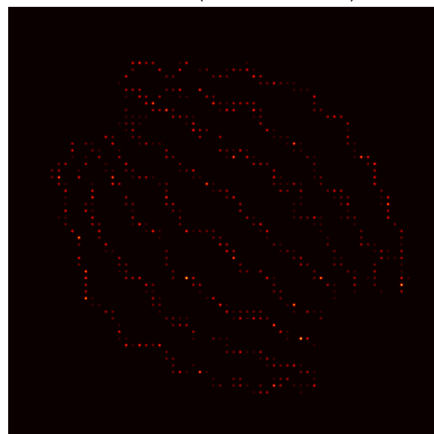


Figure 4. Version 2-4 — EPFL localization comparison.

7. Localization accuracy progression

Version	EPFL localization RMSMD	Duration(training and test)
v1 — Patch KAN	65.8 nm	10 min
v2.2 — Dense Grid	89.9 nm	23 min
v2.3 — Heatmap + Patch	52.6 nm	12 min
v2.4 — Full-Frame Dual KAN	43.7 nm	30 min

Table 8. Localization RMSMD.